Summary report on the WHO workshop on estimation of respiratory syncytial virus (RSV) disease burden based on RSV surveillance of Global Influenza Surveillance and Response System

Geneva, Switzerland, 21–23 June 2023

World Health Organization

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ISBN 978-92-4-009284-6 (electronic version)
ISBN 978-92-4-009285-3 (print version)

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Suggested citation. Summary report on the WHO workshop on estimation of respiratory syncytial virus (RSV) disease burden based on RSV surveillance of Global Influenza Surveillance and Response System, Geneva, Switzerland, 21–23 June 2023. Geneva: World Health Organization; 2024. Licence: CC BY-NC-SA 3.0 IGO.

Cataloguing-in-Publication (CIP) data. CIP data are available at https://iris.who.int/.

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Acknowledgements

We acknowdedge the contributions made during the workshop by country representatives from ministries of health and national influenza centres, colleagues from the Coronavirus and Other Respiratory Viruses Division of the United States Centers for Disease Control and Prevention (CDC), colleagues from WHO regional offices and country offices, and Dr Maria Zambon from the UK Health Security Agency (UKHSA).

This report was prepared by Mei Shang, Fernando Motta and Sergejs Nikisins from the WHO Global Influenza Programme (GIP). It was reviewed by Siddhivinayak Hirve and Wenqing Zhang, WHO GIP, as well as by the meeting Chair, Dr Maria Zambon.

Abbreviations and acronyms

BoD	burden of disease
COVID-19	coronavirus disease 2019
EQA	external quality assessment
GISRS	Global Influenza Surveillance and Response System
ICU	intensive care unit
ILI	influenza-like illness
PCR	polymerase chain reaction
RSV	respiratory syncytial virus
SARS-CoV-2	severe acute respiratory syndrome coronavirus 2
SARI	severe acute respiratory infection
WHO	World Health Organization

Background

Respiratory syncytial virus (RSV) is a leading cause of acute lower respiratory infections and hospitalization in infants and young children. The WHO RSV surveillance project, based on the Global Influenza Surveillance and Response System (GISRS) and implemented in 25 countries across all six WHO regions from 2016 to 2023, aimed to generate evidence on RSV seasonality, risk factors, spread and evolution, along with basic hospitalization proportion estimates. The Global Vaccine Alliance (Gavi) has prioritized conditional support for potential RSV vaccine products in its Vaccine Investment Strategy from 2018 to 2023. In light of ongoing efforts and the imminent licensure and availability of RSV vaccine products, it is crucial to generate country-level estimates of RSV disease burden based on surveillance in infants and young children. Such data will inform global, regional and national policies on vaccine introduction and immunization guidelines.

Scope and objectives of the workshop

The WHO RSV surveillance strategy advocates the use of a tiered approach for the estimation of RSV burden of disease (BoD) based on national priorities, capacities and resources. This workshop aimed to strengthen national capacities for implementing high-quality RSV surveillance and estimating the RSV-associated hospitalization burden in six low- and lower middle-income countries[1] based on their expression of interest, capacities and likely early adoption of RSV vaccine products once they become available.

In addition, a side meeting was conducted during the workshop that focused on practical considerations for RSV genomic sequencing.

The specific objectives of the workshop were to:

- assess national capacities to implement RSV surveillance to WHO-recommended standards;
- help countries to define the RSV-associated BoD based on available data sources, assess data suitability and identify data gaps in line with national priorities, available resources and relevance for policy/assessment of interventions;
- train participants in the use of tools to estimate RSV-associated BoD proportion and rate/incidence based on RSV sentinel surveillance data using the WHO tiered approach;
- develop country-specific protocols and analytical approaches for RSV-associated hospitalization burden estimation;
- develop a technical assistance programme for countries to provide ongoing support for RSV surveillance and BoD estimation; and
- identify the practical and operational considerations for RSV genomic sequencing.

1 Côte d'Ivoire, Egypt, India, Mongolia, Mozambique and Philippines.

Assessing respiratory syncytial virus surveillance standards in countries – identifying gaps

This session provided a progress overview and summary of achievements of the second phase of the WHO RSV surveillance project implemented in 25 countries. Discussions highlighted the opportunities offered for the surveillance of influenza and other respiratory viruses of epidemic and pandemic potential, despite the disruption consequent to the COVID-19 pandemic. The status of national compliance with WHO-recommended standards for RSV surveillance (including use of case definitions, systematic and representative sourcing of respiratory specimens, and reporting to the WHO RespiMart data platform) was also discussed.

Consideration was also given to the results of external quality assessment (EQA) of RSV detection and typing, the uploading of RSV genomic sequences to the Global Initiative on Sharing All Influenza Data (GISAID), the phylogenetic diversity of RSV-A and RSV-B subtypes, and the challenges posed by bioinformatics.

The need to integrate RSV, SARS-CoV-2 and influenza as part of pan-respiratory virus surveillance in the context of collaborative surveillance and the global architecture for health emergencies preparedness, resilience and response was highlighted. The WHO "Crafting the mosaic" tool[2] will allow countries to prioritize their surveillance objectives, strengthen existing sentinel and non-sentinel-based surveillance to ensure prompt detection of circulating respiratory viruses, and to inform public health interventions.

Table 1 provides an overview of the characteristics of the RSV surveillance programmes in the six countries that participated in the current workshop.

2 Crafting the mosaic. A framework for resilient surveillance for respiratory viruses of epidemic and pandemic potential. Geneva: World Health Organization; 2023 (https://iris.who.int/handle/10665/366689, accessed 7 April 2024).

Table 1: **Overview of RSV surveillance programmes in six countries**

Surveillance characteristic	Côte d'Ivoire	Egypt	India	Mongolia	Mozambique	Philippines
Start year	2016	2016	2015	2017	2018	2022
Number of sites	10	17	3	33	2	3
Patient source	Inpatients and outpatients	Inpatients and outpatients	Inpatients and outpatients	Inpatients and outpatients	Inpatients	Inpatients
Type of facilities	General hospitals	General and paediatric hospitals	General and paediatric hospitals	General and paediatric hospitals	General hospitals	General hospitals
Level of care	Community clinics and secondary hospitals	Tertiary hospitals	Secondary and tertiary hospitals	Community clinic, secondary and tertiary hospitals	Tertiary hospitals	Tertiary hospitals
Intensive care unit (ICU) patient enrolment	Yes	Yes	Yes	Yes	Yes	Yes
Case definition	WHO SARI and ILI	WHO SARI and ILI	WHO SARI and ILI	WHO SARI and ILI	SARI	Extended SARI
Sampling strategy	All SARI patients are enrolled along with a maximum of 3 cases daily from ILI sites	SARI sites enrol a daily range of 1–6 patients from Saturday to Friday	SARI sites enrol a minimum of 10 patients weekly (15 for ILI)	SARI sites enrol a maximum of 10 patients weekly	Sites enrol a maximum of 20 patients daily	Each site has a fixed number of patients to enrol during the course of a week in selected wards
Age group under surveillance	All ages	All ages, with a focus on children aged < 2 years	All ages	All ages, with a focus on children aged < 2 years	Children aged < 2 years	Children aged < 2 years
Specimen type	Nasopharyngeal or oropharyngeal swab	Nasopharyngeal or oropharyngeal swab	Nasopharyngeal swab	Nasopharyngeal or oropharyngeal swab	Nasopharyngeal or oropharyngeal swab	Nasopharyngeal or oropharyngeal swab
Polymerase chain reaction (PCR) capacity at site and central level	Samples are shipped on a weekly basis to a national centre for testing	Samples are shipped on a weekly basis to a national centre for testing	Samples are shipped on the same day of collection to a national centre for testing	Samples are shipped on a weekly basis to a national centre for testing	Samples are shipped on a weekly basis to a national centre for testing	Samples are shipped on a weekly basis to a national centre for testing
RSV sequencing capacity	Currently no RSV sequencing capacity at the central laboratory	National laboratory conducted 23 sequencings as of June 2023	Positive samples with a PCR cut-off value of < 25 are processed for sequencing	Currently no RSV sequencing capacity at the central laboratory	Currently no RSV sequencing capacity at the central laboratory	Currently no RSV sequencing capacity at the central laboratory
Surveillance officer onsite	One doctor oversees the surveillance at site	All sites have project staff	One dedicated laboratory staff member to overlook laboratory activities and data entry	Sentinel sites designate staff to conduct surveillance activities	All sites have project staff	One surveillance officer to ensure proper patient enrolment

Facilitators and country representatives then discussed how best to enhance national RSV surveillance and BoD estimation, with prioritized workplans for the next 10 months outlined for each country.

Côte d'Ivoire

- Complete the mapping of sentinel site performance to better understand weaknesses and propose solutions.
- Adopt the extended SARI case definition for children aged < 2 years and revise protocols to reflect this change.
- Explore and analyze existing data, and develop a manuscript for publication.
- Standardize the weekly data reporting format and upload data to RespiMart.
- Assess the quality of denominator data collected during RSV surveillance.

Egypt

- Complete the mapping of sentinel site performance.
- Update national guidelines to align with WHO recommendations.
- Review the testing algorithm and develop a rational plan taking into account reagent supply.
- Build up capacity for RSV genomic sequencing.
- Publish a paper on RSV BoD estimates using data collected from RSV surveillance.

India

- Complete the mapping of sentinel site performance and develop plans for follow-up corrective actions.
- Update protocols on RSV surveillance, including more detail on denominator data collection, as well as on monitoring and evaluation.
- Complete the data analysis plan initiated during the current workshop, conduct data analysis and summarize the results for publication.
- Enhance capacities on testing and genomic sequencing for RSV surveillance.

Mongolia

- Review current surveillance practices at 38 sites.
- Apply the extended SARI case definition for RSV surveillance and conduct exhaustive enrolment of children of younger age.
- Complete BoD estimates using data collected between October 2017 and September 2018.

Mozambique

- Narrow and clarify the case definition to be used in the sampling protocol.
- Update the sampling protocol.
- Develop policies to facilitate BoD estimation using RSV surveillance data.
- Validate laboratory processes and the testing algorithm to ensure the sustainability of reagent supply.
- Evaluate the possibility of incorporating genomic sequencing into RSV surveillance.

Philippines

- Revise the site assessment form and conduct evaluation of sentinel sites.
- Conduct enhanced training in epidemiological and laboratory activities.
- Finalize the integration of RSV surveillance into pan-respiratory pathogen surveillance.
- Optimize the RSV genomic sequencing protocols.

- Submit approval for data collection and analysis, develop information products and organize a meeting to disseminate the results.

The following key recommendations for the six countries were then developed during the workshop.

- Countries should use the extended SARI case definition for enrolling eligible hospitalized children aged < 2 years.
- Countries should consider testing at least 400 specimens annually, distributed equally among the four age groups: 0–2 months, 3–5 months, 6–12 months and 13–23 months. For estimating RSV-associated hospitalization burden, countries should test at least 1000 specimens for RSV per year.
- Countries should enrol all eligible patients OR ensure a systematic sampling approach for enrollment.
- Countries should ensure the use of an unbiased testing strategy for RSV and avoid selection bias – such as only testing for RSV if the specimen is negative for influenza or SARS-CoV-2.
- Countries should ensure the consistent and timely uploading of RSV surveillance data to the WHO RespiMart platform, synchronized with the uploading of influenza and SARS-CoV-2 surveillance data.
- Countries should develop a plan for obtaining technical assistance in assessing the performance of sentinel sites and laboratories, and for identifying and addressing gaps and challenges. Such assessment may include site visits by WHO and CDC in coordination with national counterparts.

Respiratory syncytial virus disease burden estimation

This session provided an overview of the importance of RSV surveillance in the context of the global burden of RSV infection in young children. In addition, updates were provided on the developmental pipeline for RSV vaccines and monoclonal antibodies, and on WHO efforts to increase awareness of RSV infection and preventive products among policy-makers, medical practitioners and other stakeholders. Such efforts include WHO processes for developing regulatory guidance, Strategic Advisory Group of Experts on Immunization (SAGE) recommendations, WHO prequalification and the RSV roadshow.

The WHO tiered approach to RSV disease burden estimation was presented with details of the source data required, the methods and tools to be used for the collection of numerator and denominator data, and the strengths, caveats and assumptions of each estimation method – with an emphasis placed on the collection of year-round high-quality surveillance data to avoid biased burden estimates. A specific focus was placed on the following six burden estimates:

- Proportion of Respiratory Hospital Admissions due to RSV (Tier 1.1). This approach requires data on numbers of RSV-positive cases and total tested cases, and on the sensitivity and specificity of the case definition used for surveillance.
- Proportion of Respiratory ICU Hospital Admissions and In-Hospital Deaths due to RSV (Tiers 2.2 and 2.4). This approach is based on the same amount of data and same calculation methods as Tier 1.1 except all calculations are restricted to ICU respiratory case admissions and deaths.
- Proportion of All-Cause Hospital Admissions due to RSV (Tier 1.2). In addition to Tier 1.1 data, this approach also requires the numbers of total respiratory admissions and total all-causes admissions.
- Proportion of All-Cause ICU Hospital Admissions and In-Hospital Deaths due to RSV (Tiers 2.2 and 2.4 for all-cause). This approach is based on the same amount of data and same calculation methods as Tier 1.1 except all calculations are restricted to ICU all-causes admissions and all-causes deaths.

- RSV Case Fatality Rate (Tier 2.3). This approach requires data on all RSV-positive cases and on deaths among RSV-positive cases.
- Site(s)-Specific RSV Hospitalization Rate per 100 000 Population (Tier 2.1). This approach requires the most amount of data. For the numerator calculation, the total number of RSV-positive cases, total tested cases, case definition and total number of SARI/respiratory admissions are required. For the denominator calculation, the total number of respiratory admissions, proportion of respiratory admissions out of total admissions in the sentinel site, and mid-year population estimates of the sentinel site catchment area are needed.

Côte d'Ivoire

- Tiers 1.1 and 1.2: the proportional contribution of RSV-associated illness among SARI and all-cause hospitalizations at selected sentinel sites during 2016–2019 among the following age groups: 0–2, 3–5, 6–11, 12–23 and 24–59 months.
- Tier 2.1: the annual/mean annual number and rates of RSV-associated SARI hospitalization at selected sentinel sites during 2016–2019 among the following age groups: 0–2, 3–5, 6–11, 12–23 and 24–59 months.

Egypt

- Tier 2.1: the mean annual number and rates of RSV-associated SARI and RSV-associated hospitalization at selected sentinel sites during 2022–2023 among the following age groups: 0–5, 6–11 and 12–23 months (pending confirmation of sufficient sample sizes in each age group).

India

- Tiers 1.1 and 1.2: the proportional contribution of RSV-associated illness among SARI/all-respiratory/all-cause hospitalizations at selected sentinel sites during 2021–2023 among the following age groups: 0–2, 3–5, 6–11, 12–23 and 24–59 months.
- Tiers 2.2 and 2.4: the proportional contribution of RSV-associated illness among SARI/all-respiratory/all-cause ICU admissions/in-hospital deaths at selected sentinel sites during 2021–2023 among the following age groups: 0–2, 3–5, 6–11, 12–23 and 24–59 months.

Mongolia

- Tiers 1.1 and 1.2: the proportional contribution of RSV-associated illness among SARI/all-respiratory/all-cause hospitalizations at 8 sentinel sites in Ulaanbaatar during October 2017–September 2018 among the following age groups: 0–2, 3–5, 6–11, 12–23 and 24–59 months.
- Tiers 2.2 and 2.4: the proportional contribution of RSV-associated illness among SARI/all-respiratory/all-cause ICU admissions/in-hospital deaths at 8 sentinel sites in Ulaanbaatar during October 2017–September 2018 among the following age groups: 0–2, 3–5, 6–11, 12–23 and 24–59 months.
- Tier 2.3: the mortality/ICU admission/ventilatory support proportion among RSV-positive patients hospitalized at 8 sentinel sites in Ulaanbaatar during October 2017–September 2018 among the following age groups: 0–2, 3–5, 6–11, 12–23 and 24–59 months.

Mozambique

- Tiers 1.1 and 1.2: the proportional contribution of RSV-associated illness among SARI/all-respiratory/all-cause hospitalizations at selected sentinel sites prospectively from 2023 among the following age groups: 0–2, 3–5, 6–11 and 12–23 months.
- Tier 2.3: the mortality/ICU admission/ventilatory support proportion among RSV-positive patients hospitalized at selected sentinel sites during 2018–2022 (or prospectively 2023–2024 depending on data availability) among the following age groups: 0–2, 3–5, 6–11 and 12–23 months.

Philippines

- Tiers 1.1 and 1.2: the proportional contribution of RSV-associated illness among expanded SARI and all-cause admission at selected sentinel sites during May 2022–April 2024 among the following age groups: 0–2, 3–5, 6–11 and 12–23 months.
- Tiers 2.2 and 2.4: the proportional contribution of RSV-associated illness among expanded SARI/all-respiratory/all-cause ICU admissions/in-hospital deaths at selected sentinel sites during May 2022–April 2024 among the following age groups: 0–2, 3–5, 6–11 and 12–23 months.
- Tier 2.3: the mortality/ICU admission/ventilatory support proportion among RSV-positive patients hospitalized at selected sentinel sites during May 2022–April 2024 among the following age groups: 0–2, 3–5, 6–11 and 12–23 months.
- Tier 2.1: the annual/mean annual number and rates of RSV-associated expanded SARI/all-respiratory hospitalization at selected sentinel sites during May 2022–April 2024 among the following age groups: 0–2, 3–5, 6–11 and 12–23 months.

Other surveillance-related recommendations and actions

Countries identified the following areas for technical and logistics support from WHO/CDC for RSV surveillance and BoD estimation:

- WHO to create a SharePoint and schedule quarterly calls to facilitate the setting up of an international "community of practice" as a forum for sharing experiences and progress, and promoting collaboration.
- WHO to create a dashboard accessible to participating countries to allow for the monitoring of surveillance process and output indicators.
- WHO to create a resource hub that includes tools for site assessment, bioinformatics learning tools and training courses on epidemiology.
- WHO to develop a tool for right-sizing RSV surveillance.
- WHO to facilitate the use of surveillance and disease burden data to inform public interventions and policy, including through the provision of WHO assistance in surveillance data analysis, reporting and dissemination.
- WHO to develop country factsheets/other information products to inform policy-makers and other stakeholders, and to explore and adopt information product approaches used by other successful paediatric immunization programmes for use in RSV prevention.
- WHO to organize a virtual training workshop on RSV data management for data managers in the six countries.
- WHO to facilitate the availability and distribution of CDC multiplex assays for RSV typing through the CDC International Reagent Resource.
- WHO to implement the next round of EQA for RSV detection and typing in 2023.

Side meeting on practical considerations for respiratory syncytial virus genomic sequencing

This side meeting was attended by representatives of the RSV laboratories and external subject matter experts, and focused on the objectives of RSV molecular surveillance, sampling strategy, laboratory considerations for genomic sequencing, and sharing of sequence data. The immediate priorities identified for RSV genomic surveillance included setting standards, developing bioinformatics pipelines for genome assembly and annotation, standardization of lineage classification, and right-sizing sequencing to characterize global RSV diversity. Discussion topics regarding post-sequencing analysis included available analysis pipelines, and the desired outcomes for RSV genomic surveillance in the context of challenges such as required computational infrastructure and bioinformatics expertise. Further issues highlighted included the need for a grading system for RSV sequencing EQA, the role of metadata, and ensuring the timeliness of sequence data sharing. A proposal was presented on the integration of RSV lineage classification into global genomic surveillance. The proposal covered key points such as the use of complete genomes for molecular classification and of the ReSVidex tool for lineage classification. The significance of careful storage and year-round collection of RSV samples sourced from sentinel surveillance, and of quality assurance of the generated sequences, was emphasized. Other recommendations included the need for RSV programmes to link genotype to phenotype, to link funding to the public release of data and to encourage countries to share the results of their work.

Specific action points identified during discussion included drafting of a guidance resource on RSV genomic sequencing considerations, coordinating with the group currently working on RSV lineage classification, and exploring formal modelling options for the sampling strategy in collaboration with partners.

Annex 1: Meeting agenda

Chair: Maria Zambon

Facilitators for RSV BoD pilot countries

Côte d'Ivoire – Julie Thwing
Mozambique – Laura Wright
Egypt – Eva Leidman
India – Siddhi Hirve
Mongolia – Mei Shang
Philippines – Stefano Tempia

21 June 2023

09:00–09:20	Welcome, introductions, objectives and expected outcomes of the workshop, disclosure of interests, and selection of Chair	Wenqing Zhang
Session 1: RSV surveillance – current state of play		
09:20–09:40	WHO RSV surveillance project – overview, lessons learned and priorities	Siddhi Hirve
09:40–10:00	Data in context – biopsy survey (epi), data quality, plan of analysis and early findings	Mei Shang
10:00–10:20	Data in context – biopsy survey (lab) and RSV genomic data	Fernando Motta
10:20–10:30	Q&A	
10:30–11:00	*Group photo, coffee break, participants split into BoD and Genomics side meetings*	
11:00–13:00	RSV surveillance standards – addressing gaps (sentinel site and laboratory) • 1:1 discussion with countries	Country + Facilitator
Session 2: RSV disease burden		
14:00–14:15	RSV burden – why now?	Danny Feikin
14:15–16:00	RSV burden • Tiered approach • Proportions – data needs, sources and methods • Incidence – data needs, sources and methods	Stefano Tempia
16:20–18:00	Plenary feedback from 1:1 discussion with countries – addressing gaps in surveillance standards, and availability of BoD related data	Country representatives
18:00–20:00	Reception	

22 June 2023

Session 3: Developing country-specific RSV BoD protocols		
08:30–09:10	Plenary feedback from 1:1 discussion with countries – addressing gaps in surveillance standards	Côte d'Ivoire, Mozambique, Mongolia
09:10–09:20	• Briefing for the day • Country-specific BoD protocol	Siddhi Hirve Stefano Tempia
09:20–12:00	RSV BoD priority estimates, data needs, sources, collection methodology, quality control, plan of analysis, dissemination and use of evidence • 1:1 discussion with countries	Country + Facilitator
12:00–12:30	5-minute update on discussions from each country	Country representatives
13:30–13:50	RSV data upload in RespiMart • Q&A	Bikram Maharjan
13:50–15:30	Country + facilitator to draft BoD protocol	
15:30–16:00	*Coffee break – Genomics side meeting participants regroup with BoD side meeting participants*	
16:00–17:00	Plenary feedback – operational considerations for RSV sequencing • Discussion	Thomas Williams

23 June 2023

Session 4: Outcomes and actions		
09:00–09:10	Briefing for the day	Siddhi Hirve Stefano Tempia
09:10–12:30	Technical assistance plan – define surveillance gaps, solutions, support from WHO, workplan and timelines 1:1 discussion with countries	Country + Facilitator
13:30–14:00	Technical assistance – how best to make it work? Moderated discussion	Eva Leidman Pavana Murthy
14:00–14:30	RSV BoD outputs – dissemination and advocacy Moderated discussion	Stefano Tempia Eva Leidman
15:00–16:20 16:20–16:30	Presentation on Mentorship plan Next steps	Country + Facilitator Siddhi Hirve
16:30–17:00	Closing remarks	Maria Zambon Wenqing Zhang

Annex 2: List of participants

Participants – in person

Sumit Dutt Bhardwaj	National Institute of Virology Pune, India
Daouda Coulibaly	Institut National d'Hygiène Publique Abidjan, Côte d'Ivoire
Ola Salah Deghedy	General Department of Epidemiology and Surveillance, Ministry of Health and Population Cairo, Egypt
Steven Edwards	Victorian Infectious Diseases Reference Laboratory Melbourne, Australia
Shonkhuuz Enkhtur	National Center for Maternal and Child Health Ulaanbaatar, Mongolia
Hervé Kadjo	Institut Pasteur de Côte d'Ivoire Abidjan, Côte d'Ivoire
Eva Leidman	Centers for Disease Control and Prevention Atlanta, United States of America
Jonjee C. Morin	Research Institute for Tropical Medicine Manila, Philippines
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Devon Rayo Pacial	Epidemiology Bureau, Department of Health Manila, Philippines
Varsha Potdar	National Institute of Virology Pune, India
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